

Your Cyclotomic LHL Guide

Joint work with Katharina Boudgoust

Sasha Lapiha, Royal Holloway University of London

Building your lattice primitive over rings?

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Trying to get your security theorem right?

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Trying to get your security theorem right?

And now you come across a Leftover Hash Lemma (LHL)?

When is it an LHL?

$$(A, A \cdot x \bmod q) \sim (A, \mathcal{U}(\mathcal{R}_q^n))$$

where

$$A \leftarrow \mathcal{U}(\mathcal{R}_q^{n \times m}), \quad x \leftarrow \chi$$

and χ is usually a short distribution over $\mathcal{R}^m$ or $\mathcal{R}_q^m$

What do you do?

LUNA: Quasi-Optimally Succinct Designated-Verifier Zero-Knowledge Arguments from Lattices

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Plug-and-play sanitization for TFHE

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Making NTRUEncrypt and NTRUSig Worst-Case Problems over

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Hardness of Entropic Module-LWE

Hao Lin^{1,2}, Mingqiang Wang^{1,2}, Jincheng Zhuang^{2,3}, and Yang Wang^{1,2}

Discrete Gaussians Modulo Sub-Lattices: New Leftover Hash Lemmas for Discrete Gaussians

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Efficient Public Key Encryption Based on Ideal Lattices

Damien Stehlé^{1,2}, Ron Steinfeld², Keisuke Tanaka³, and Keita Xagawa³

On the Ring-LWE and Polynomial-LWE Problems

Miruna Rosca^{1,2}, Damien Stehlé¹, and Alexandre Wallet¹

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² Bitdefender, Romania

Partitioning via Non-linear Polynomial Functions: More Compact IBEs from Ideal Lattices and Bilinear Maps

Shuichi Katsumata^{1,2} and Shota Yamada^{2(✉)}

GENERALIZED COMPACT KNAPSACKS, CYCLIC LATTICES, AND EFFICIENT ONE-WAY FUNCTIONS

DANIELE MICCIANCIO

On the Hardness of Module Learning With Errors with Short Distributions

Katharina Boudgoust¹, Corentin Jeudy^{2,3}, Adeline Roux-Langlois⁴, and Weiqiang Wen⁵

A Toolkit for Ring-LWE Cryptography

Vadim Lyubashevsky* Chris Peikert† Oded Regev‡

May 16, 2013

Rounding in the Rings

Feng-Hao Liu* Zhedong Wang†‡

Dana Dachman-Soled, Huijing Gong*, Mukul Kulkarni, and Aria Shahverdi

Towards a Ring Analogue of the Leftover Hash Lemma

and more...

Or you read the ring LHL Guide!

Leftover Hash Lemma(s) Over Cyclotomic Rings

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- Leakage?????

We cover it all !

- What is my ring/my splitting? - all cyclotomic rings at your service!
 - Renyi or Statistical? - SD and RD for all statements
 - Matrix distribution? - Uniform, $[I \mid A]$, and “invertible coefficient” matrices.
 - Lattice embedding? - statements in both embeddings
 - God what if it is not a Gaussian? - Guide on Ternary / Centered Binomial / etc in LHL
 - Leakage????? - Entropic, Linear and Noisy Linear leakage!
- + most classic proof techniques in the area spelled out

Be the Pro of your LHL Problems!

Paper: “Leftover Hash Lemma(s) over Cyclotomic Rings”.
Soon to appear on eprint.

If we didn't quite cover something. Please let us know!

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